

- **LLM Inference Engine**

A from-scratch LLM inference engine in PyTorch (KV cache, batching, paged memory)

[Github](#)

- Built a PyTorch LLM inference engine for GPT-2 and Mistral with KV caching, continuous batching, paged memory, and async serving.
- Achieved 2.22x decode throughput and cut CUDA time 46.6% with a preallocated KV cache on an A100.
- Scaled GPT-2 124M from 155 to 18,346 tokens/second, a 118.3x gain at batch size 512 on an A100.
- Cut peak KV-cache memory up to 35.9%, delivered 3.2x throughput at 64 concurrent requests, and added 122 tests.